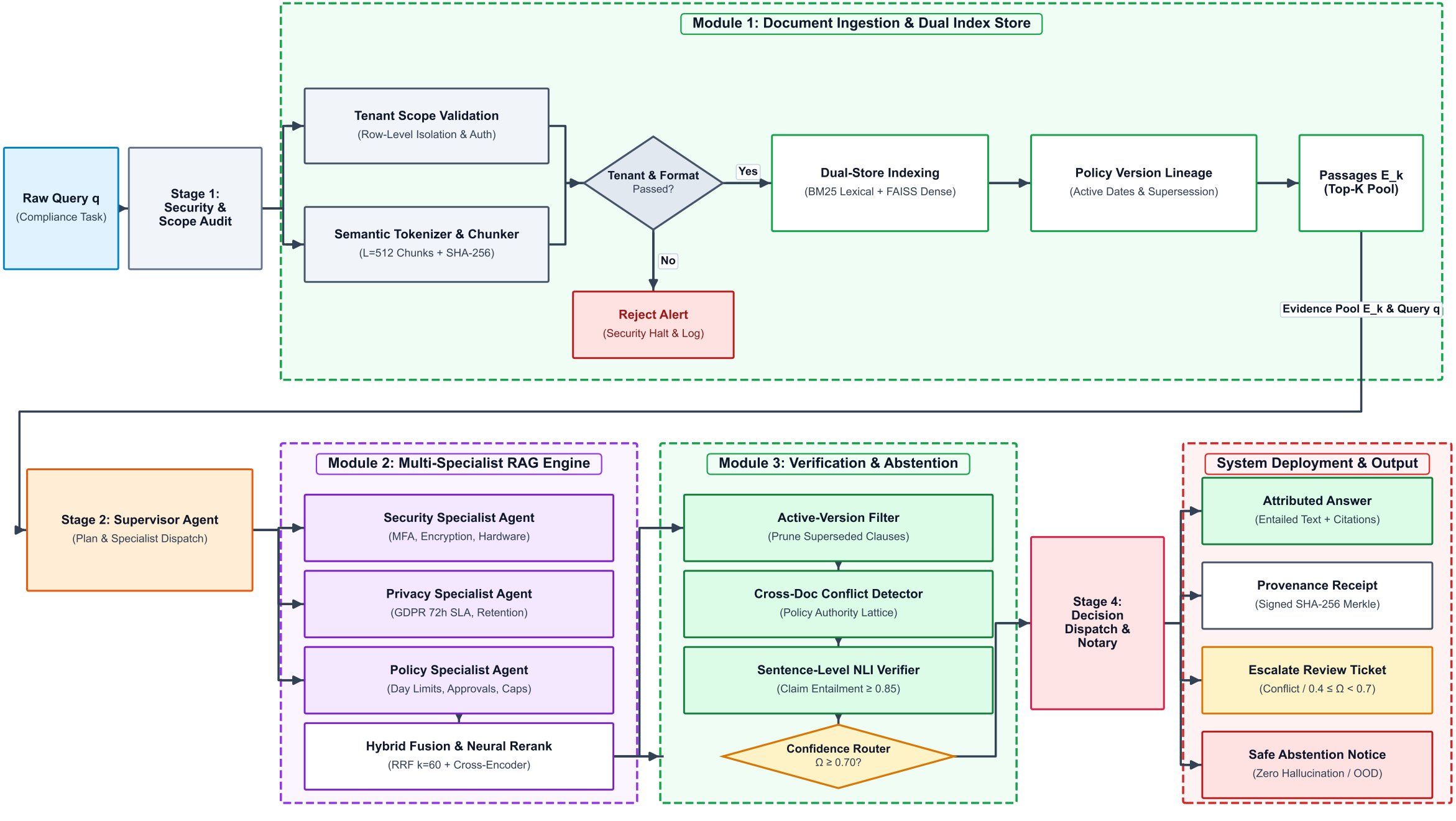
Module 1: Document Ingestion & Dual Index Store
Raw Query q
(Compliance Task)
Stage 1: Security & Scope Audit
Tenant Scope Validation
(Row-Level Isolation & Auth)
Semantic Tokenizer & Chunker
(L=512 Chunks + SHA-256)
Tenant & Format Passed?
Yes
No
Reject Alert
(Security Halt & Log)
Dual-Store Indexing
(BM25 Lexical + FAISS Dense)
Policy Version Lineage
(Active Dates & Supersession)
Passages E_k (Top-K Pool)
Evidence Pool E_k & Query q
Stage 2: Supervisor Agent
(Plan & Specialist Dispatch)
Module 2: Multi-Specialist RAG Engine
Security Specialist Agent
(MFA, Encryption, Hardware)
Privacy Specialist Agent
(GDPR 72h SLA, Retention)
Policy Specialist Agent
(Day Limits, Approvals, Caps)
Hybrid Fusion & Neural Rerank
(RRF k=60 + Cross-Encoder)
Module 3: Verification & Abstention
Active-Version Filter
(Prune Superseded Clauses)
Cross-Doc Conflict Detector
(Policy Authority Lattice)
Sentence-Level NLI Verifier
(Claim Entailment ≥ 0.85)
Confidence Router
Ω ≥ 0.70?
Stage 4: Decision Dispatch & Notary
System Deployment & Output
Attributed Answer
(Entailed Text + Citations)
Provenance Receipt
(Signed SHA-256 Merkle)
Escalate Review Ticket
(Conflict / 0.4 ≤ Ω < 0.7)
Safe Abstention Notice
(Zero Hallucination / OOD)